

Program 9

Organize Kubernetes Pods using Labels and Namespaces

Steps

1. Create 2 files : namespaces.yaml and pods.yaml

Inside namespaces.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Inside pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
labels:
  app: frontend
  env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
labels:
  app: backend
  env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
labels:
  app: frontend
  env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
labels:
  app: backend
  env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

2. Apply both .yaml files

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

3. Get pods based on namespaces

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl get pods -n prod
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          14s
frontend-pod  0/1     ContainerCreating   0          14s

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          22s
frontend-pod  1/1     Running   0          22s
```

4. Get pods based on Labels

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          32s

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          44s
```

5. Delete all pods

kubectl delete pods -all

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl delete pods -n dev --all
pod "backend-pod" deleted
pod "frontend-pod" deleted

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program9
$ kubectl delete pods -n prod --all
pod "backend-pod" deleted
pod "frontend-pod" deleted
```